

LONGERENONG, Australia

WMO: 958350

Lat: 36.6722S

Lon: 142.2992E

Elev: 436

StdP: 14.47

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.2	34.2	23.0	17.6	74.1	26.9	21.2	71.2	28.9	50.0	25.5	53.3	3.7	240	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	100.1	65.7	95.6	64.1	91.7	63.5	70.7	84.0	68.2	84.7	66.2	83.6	13.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	102.2	73.3	63.8	90.0	71.5	60.8	80.9	70.9	34.9	83.1	32.6	83.7	31.1	83.4	97.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	22.7	20.1	DB	28.1	108.7	1.9	3.1	26.7	110.9	25.6	112.8	24.6	114.5	23.3	116.8
			WB	27.8	77.2	1.8	9.3	26.5	83.8	25.5	89.3	24.5	94.5	23.2	101.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	71.4	71.0	65.7	59.1	52.9	48.5	47.6	48.5	51.7	56.1	62.7	67.0	(6)
	DBStd	10.61	8.14	7.67	7.51	6.21	4.75	3.77	3.45	3.98	5.17	6.97	8.04	8.20	(7)
	HDD50	314	0	0	0	3	22	71	89	77	35	15	2	0	(8)
	HDD65	3156	25	26	84	197	375	495	539	513	399	292	139	72	(9)
	CDD50	3395	663	587	485	276	112	26	15	30	88	204	383	526	(10)
	CDD65	761	223	193	104	20	0	0	0	0	1	16	70	134	(11)
	CDH74	11144	3246	2472	1335	289	8	0	0	0	35	388	1198	2173	(12)
	CDH80	5505	1769	1243	583	76	0	0	0	0	4	140	550	1140	(13)
(14) Wind	WSAvg	9.8	11.2	10.9	10.5	8.9	8.8	8.3	9.1	9.2	9.4	9.7	10.3	10.9	(14)
(15) Precipitation	PrecAvg	16.6	1.0	0.8	0.8	1.1	1.8	1.6	1.9	1.9	1.8	1.8	1.3	0.9	(15)
	PrecMax	31.9	3.9	6.1	3.9	4.6	5.6	5.7	3.9	3.7	4.8	6.8	3.4	2.8	(16)
	PrecMin	7.5	0.0	0.0	0.0	0.0	0.2	0.3	0.5	0.2	0.4	0.1	0.0	0.0	(17)
	PrecStd	4.7	0.9	1.1	0.8	1.0	1.2	1.1	0.9	0.8	1.1	1.3	0.9	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.4	103.4	98.4	87.7	74.5	63.7	62.0	69.4	78.8	90.9	97.7	105.1	(19)
		MCWB	67.2	68.1	64.5	62.8	57.0	53.9	52.2	55.2	57.2	59.4	65.8	64.7	(20)
	2%	DB	100.8	97.0	91.9	81.3	69.2	60.1	58.6	63.7	72.1	84.1	92.0	97.8	(21)
		MCWB	66.0	65.5	63.7	59.4	55.7	53.0	51.4	52.3	56.8	58.4	62.8	63.8	(22)
	5%	DB	95.0	92.6	86.6	76.7	65.3	58.0	56.5	59.9	67.1	78.0	86.8	92.0	(23)
		MCWB	64.1	65.2	62.3	58.3	54.6	52.1	50.6	51.4	54.7	56.9	62.0	62.4	(24)
	10%	DB	90.0	87.8	81.4	72.4	62.2	56.1	54.6	57.3	62.9	72.1	80.9	86.2	(25)
		MCWB	63.4	63.7	60.9	56.6	54.0	51.0	49.7	50.3	53.4	55.3	60.0	60.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.0	77.9	68.9	64.7	61.3	57.2	55.3	56.9	61.6	64.6	72.5	72.8	(27)
		MCDB	85.9	80.3	83.7	78.7	64.9	59.3	57.5	63.8	71.0	79.6	77.7	87.3	(28)
	2%	WB	69.9	71.1	66.5	62.0	58.3	54.6	53.1	54.3	58.4	61.0	66.4	67.1	(29)
		MCDB	85.4	84.5	82.3	72.4	64.6	58.0	56.4	60.6	67.1	74.9	80.7	84.0	(30)
	5%	WB	67.4	68.2	64.6	60.1	56.5	53.1	51.6	52.5	56.2	58.8	63.9	64.7	(31)
		MCDB	88.2	81.9	80.4	71.7	62.4	56.5	54.8	58.3	64.6	72.4	79.7	83.4	(32)
	10%	WB	65.3	66.0	62.7	58.3	54.9	51.8	50.3	51.0	54.1	56.8	61.6	62.7	(33)
		MCDB	86.1	81.9	78.9	70.0	60.7	55.0	53.5	56.4	61.8	70.0	76.3	82.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	29.5	27.5	25.5	23.6	18.6	16.0	15.1	18.2	21.8	26.5	28.1	29.3	(35)
		MCDBR	36.2	33.5	31.5	29.5	23.4	18.2	17.5	22.7	28.4	36.8	36.5	37.6	(36)
		MCWBR	11.9	10.9	11.0	13.0	12.4	12.0	11.4	13.2	14.8	15.1	12.9	12.2	(37)
	5% WB	MCDBR	28.7	25.3	26.7	24.1	18.3	15.3	15.1	19.8	24.1	28.9	29.9	29.1	(38)
		MCWBR	12.4	11.3	11.2	12.1	11.1	11.3	11.1	12.6	14.3	14.0	14.2	12.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.338	0.333	0.316	0.309	0.301	0.292	0.292	0.304	0.322	0.331	0.337	0.330	(40)
	taud		2.496	2.520	2.566	2.565	2.574	2.594	2.591	2.534	2.472	2.457	2.472	2.511	(41)
	Ebn at Noon		316	310	304	287	271	265	272	284	297	307	314	320	(42)
	Edh at Noon		36	34	30	26	23	21	22	27	32	36	37	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2443	2172	1723	1222	823	655	705	995	1377	1836	2165	2399	(44)
(45)	RadStd		125	103	78	64	39	49	42	63	98	131	136	115	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (27 neighbors)	+0.77	N/A	N/A	+1.55	N/A	N/A	N/A	-157	+250	+106	(47)

Nomenclature: See separate page